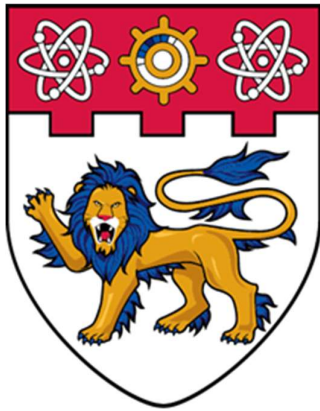




**NANYANG
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SC2006 Software Engineering
Lab 4 Deliverables
Tutorial Group : SCE2

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1. Black Box Testing	3
I. AuthController	3
II. InputController	3
III. SaveltineraryController	3
IV. Black Box testing techniques	3
V. Test Cases and Results	4
2. White Box Testing	9
1. Login	9
1.1 Control Flow Graph	9
1.2 Basic Path Testing	9
1.3 Test Cases and Results	10
2. Validate Location	12
2.1 Control Flow Graph	12
2.2 Basic Path Testing	13
2.3 Test Cases and Results	13

1. Black Box Testing

I. AuthController

The AuthController class manages user login, user registration and password reset within the app.

II. InputController

The InputController class manages Location Validation and the Search Location Auto-Suggest feature of the app.

III. SaveltineraryController

The SaveltineraryController manages the Save Route feature of the app.

IV. Black Box testing techniques

Equivalence Class Testing

The main idea is to divide all possible input data into a few groups, or "classes," where the system is expected to behave similarly for all inputs within the same class.

Instead of testing every single possible input (which is often impossible), you just pick **one representative value** from each class. The assumption is: if it works for one, it will work for all others in that class.

Boundary Value Testing

This technique is a natural next step after Equivalence Class Testing. It's based on the common experience that many bugs are found right at the **edges or "boundaries"** of the input ranges.

V. Test Cases and Results

a) Login

Input Parameters: Email, Password

No.	Test Input	Expected Output	Actual Output	Pass?
1	(Valid) Email : "user1@gmail.com" (Valid) Password : "Password"	Successful login	Successful login	Yes
2	(Invalid) Email : " (Valid) Password : "Password"	Login failed, system notify "invalid login credentials, please try again"	Login failed, system notify "invalid login credentials, please try again"	Yes
3	(Valid) Email : "user1@gmail.com" (Invalid) Password : "	Login failed, system notify "invalid login credentials, please try again"	Login failed, system notify "invalid login credentials, please try again"	Yes
4	(Invalid) Email : "user1" (Valid) Password : "Password"	Login failed, system notify "invalid login credentials, please try again"	Login failed, system notify "invalid login credentials, please try again"	Yes
5	(Valid) Email : "user1@gmail.com" (Invalid) Password : "Pass"	Login failed, system notify "invalid login credentials, please try again"	Login failed, system notify "invalid login credentials, please try again"	Yes

b) Register

Input Parameters: First Name, Last Name, Email, Address, Contact Number, Password, Confirm Password

No.	Test Input	Expected Output	Actual Output	Pass?
1	(All valid inputs) First Name : "Xiao" Last Name : "Ming" Email : " user1@gmail.com " Address : " 50 Nanyang Ave, Singapore 639798 " Contact Number : "99996666" Password : "Password@1" Confirm Password : "Password@1"	Successful signup	Successful signup	Yes

2	(All Valid Inputs except First Name) First Name : “ “	System Notify “ Please input all fields.”	System Notify “ Please input all fields.”	Yes
3	(All Valid Inputs except Last Name) Last Name : “ “	System Notify “ Please input all fields.”	System Notify “ Please input all fields.”	Yes
4	(All Valid Inputs except Email) Email : “ “	System Notify “ Please input all fields.”	System Notify “ Please input all fields.”	Yes
5	(All Valid Inputs except Address) Address : “ “	System Notify “ Please input all fields.”	System Notify “ Please input all fields.”	Yes
6	(All Valid Inputs except Contact Number) Contact Number : “ “	System Notify “ Please input all fields.”	System Notify “ Please input all fields.”	Yes
7	(All Valid Inputs except Password) Password : “ “	System Notify “ Please input all fields.”	System Notify “ Please input all fields.”	Yes
8	(All Valid Inputs except Password) Password : “pass“	System Notify “ Password must be at least 8 characters”	System Notify “ Password must be at least 8 characters”	Yes
9	(All Valid Inputs except Password) Password : “password1“	System Notify “ Password must include at least 1 special character”	System Notify “ Password must include at least 1 special character”	Yes
10	(All Valid Inputs except Password) Password : “password@“	System Notify “ Password must include at least 1 numerical digit ”	System Notify “ Password must include at least 1 numerical digit ”	Yes
11	(All Valid Inputs except Confirm Password) Confirm Password : “ “	System Notify “ Please input all fields.”	System Notify “ Please input all fields.”	Yes
12	(All Valid Inputs except Confirm Password) Password “ Password@1 ” Confirm Password : “ Password “	System Notify “ Password and Confirm Password must match”	System Notify “ Password and Confirm Password must match”	Yes

c) Password Reset

Input Parameters: New Password, Confirm New Password

No.	Test Input	Expected Output	Actual Output	Pass?
1	(Valid) Password : "Password@2" (Valid) Confirm Password : "Password@2"	Successful Reset	Successful Reset	Yes
2	(Invalid) Password : " " (Invalid) Confirm Password : " "	Reset failed, system notify "Please input all fields"	Reset failed, system notify "Please input all fields"	Yes
3	(Invalid) Password : " " (Valid) Confirm Password : "Password@2"	Reset failed, system notify "Please input all fields"	Reset failed, system notify "Please input all fields"	Yes
4	(Valid) Password : "Password@2" (Invalid) Confirm Password : " "	Reset failed, system notify "Please input all fields"	Reset failed, system notify "Please input all fields"	Yes
5	(Valid) Password : "Password@2" (Invalid) Confirm Password : "Password@1 "	System Notify " Password and Confirm Password must match"	System Notify " Password and Confirm Password must match"	Yes
6	(Invalid) Password : "pass"	System Notify " Password must be at least 8 characters"	System Notify " Password must be at least 8 characters"	Yes
7	(Invalid) Password : "password1"	System Notify " Password must include at least 1 special character"	System Notify " Password must include at least 1 special character"	Yes
8	(Invalid) Password : "password@"	System Notify " Password must include at least 1 numerical digit "	System Notify " Password must include at least 1 numerical digit "	Yes

d) Validate Location

Input Parameters: Start Location, End Location

No.	Test Input	Expected Output	Actual Output	Pass?
1	(Valid) Start Location : "50 Nanyang Ave, Singapore 639798" (Valid) End Location : " 301 Boon Lay Wy, Singapore 649846"	Location Validation Successful	Location Validation Successful	Yes
2	(Valid) Start Location : "50 Nanyang Ave, Singapore 639798" (Invalid) End Location : " "	System Notify "Invalid location. Please enter a valid Singapore location and try again."	System Notify "Invalid location. Please enter a valid Singapore location and try again."	Yes
3	(Invalid) Start Location : " " (Valid) End Location : " 301 Boon Lay Wy, Singapore 649846"	System Notify "Invalid location. Please enter a valid Singapore location and try again."	System Notify "Invalid location. Please enter a valid Singapore location and try again."	Yes
4	(Invalid) Start Location : " " (Invalid) End Location : " "	System Notify "Invalid location. Please enter a valid Singapore location and try again."	System Notify "Invalid location. Please enter a valid Singapore location and try again."	Yes
5	(Valid) Start Location : "50 Nanyang Ave, Singapore 639798" (Invalid) End Location : "Orchard"	End Location = "Orchard Road" Location Validation Successful	End Location = "Orchard Road" Location Validation Successful	Yes
6	(Invalid) Start Location : "Orchard" (Valid) End Location : " 301 Boon Lay Wy, Singapore 649846"	Start Location = "Orchard Road" Location Validation Successful	Start Location = "Orchard Road" Location Validation Successful	Yes

e) Search Location Auto-Suggest

Input Parameters: Start Location, End Location

No.	Test Input	Expected Output	Actual Output	Pass?
1	(Valid) Start Location : "Ochard"	Dropdown below Start Location search bar "Orchard Road, Orchard Road Plaza Singapura, Orchard Central"	Dropdown below Start Location search bar "Orchard Road, Orchard Road Plaza Singapura, Orchard Central"	Yes
2	(Valid) End Location : "Ochard"	Dropdown below End Location search bar "Orchard Road, Orchard Road Plaza Singapura, Orchard Central"	Dropdown below End Location search bar "Orchard Road, Orchard Road Plaza Singapura, Orchard Central"	Yes
3	(Invalid) Start Location : "!@#"	Dropdown below Start Location search bar will be Empty	Dropdown below Start Location search bar will be Empty	Yes
4	(Invalid) End Location: "!@#"	Dropdown below End Location search bar will be Empty	Dropdown below End Location search bar will be Empty	Yes

f) Saved Route

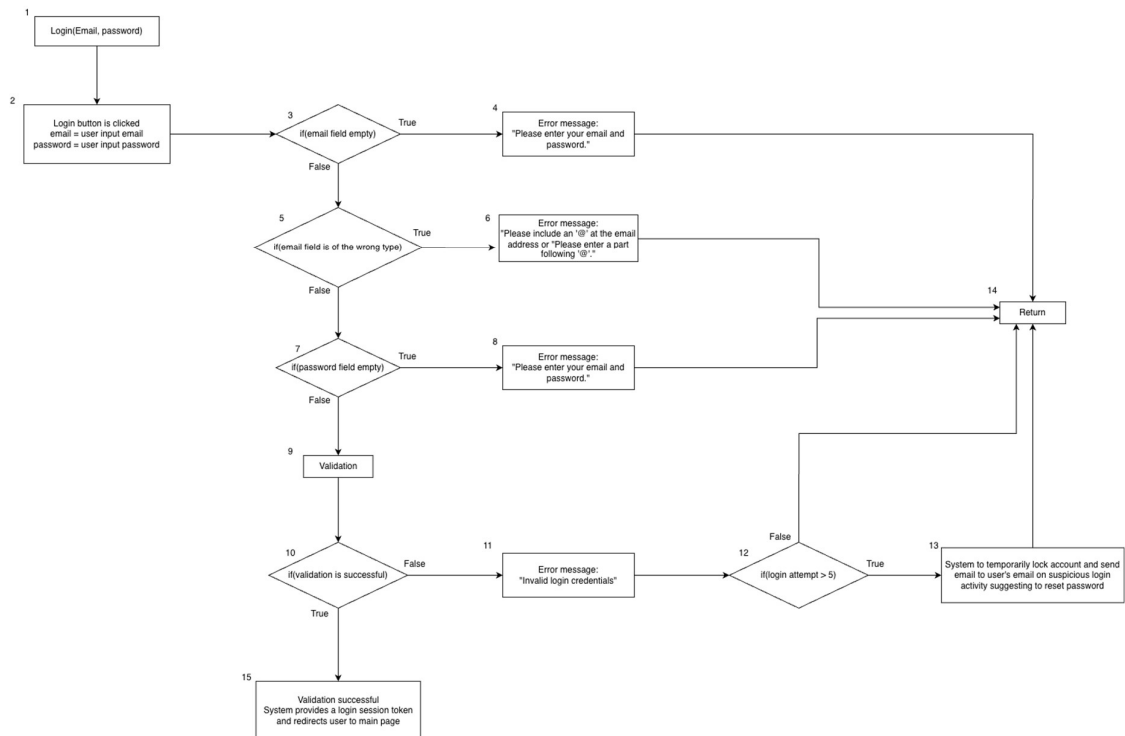
Input Parameters: Save Itinerary Button

No.	Test Input	Expected Output	Actual Output	Pass?
1	(Valid) Save Itinerary Button == True (Valid) User is logged in	Itinerary Successfully saved	Itinerary Successfully saved	Yes
2	(Valid) Save Itinerary Button == True (Invalid) User is not logged in	Login Prompt will appear	Login Prompt will appear	Yes

2. White Box Testing

1. Login

1.1 Control Flow Graph



1.2 Basic Path Testing

Cyclomatic Complexity = | decision points | + 1 = 5 + 1 = 6

Basis paths

1. Baseline Path: 1, 2, 3, 5, 7, 9, 10, 15
2. Basis Path 1: 1, 2, 3, 4, 14
3. Basis Path 2: 1, 2, 3, 5, 6, 14
4. Basis Path 3: 1, 2, 3, 5, 7, 8, 14
5. Basis Path 4: 1, 2, 3, 5, 7, 9, 10, 11, 12, 14
6. Basis Path 5: 1, 2, 3, 5, 7, 9, 10, 11, 12, 13, 14

Explanation of basis paths

Baseline Path: User to input login credentials and clicks the login button. System checks for empty email or password fields. No empty fields detected, system to check and validate account details in accordance to input and validation. Validation is successful and brings the user to the main page.

Path 1: User to input login credentials and clicks the login button. System checks and detects empty email field. System to display error message to prompt user to fill in email field. Returns the user to the login page.

Path 2: User to input login credentials and clicks the login button. System checks and detects the email field is of the wrong type. System to display error message to prompt user to check and re-enter email field. Returns the user to the login page.

Path 3: User to input login credentials and clicks the login button. System checks and detects the password field is empty. System to display error message to prompt user to fill in password field. Returns the user to the login page.

Path 4: User to input login credentials and clicks the login button. System checks for empty email or password fields. No empty fields detected, system to check and validate account details in accordance to input and validation. Validation is unsuccessful, system to return user to login page if login attempt is less than 5 times.

Path 5: User to input login credentials and clicks the login button. System checks for empty email or password fields. No empty fields detected, system to check and validate account details in accordance to input and validation. Validation is unsuccessful and login attempt is more than 5 times, system to temporarily lock account and email user on suspicious activity suggesting to reset password.

1.3 Test Cases and Results

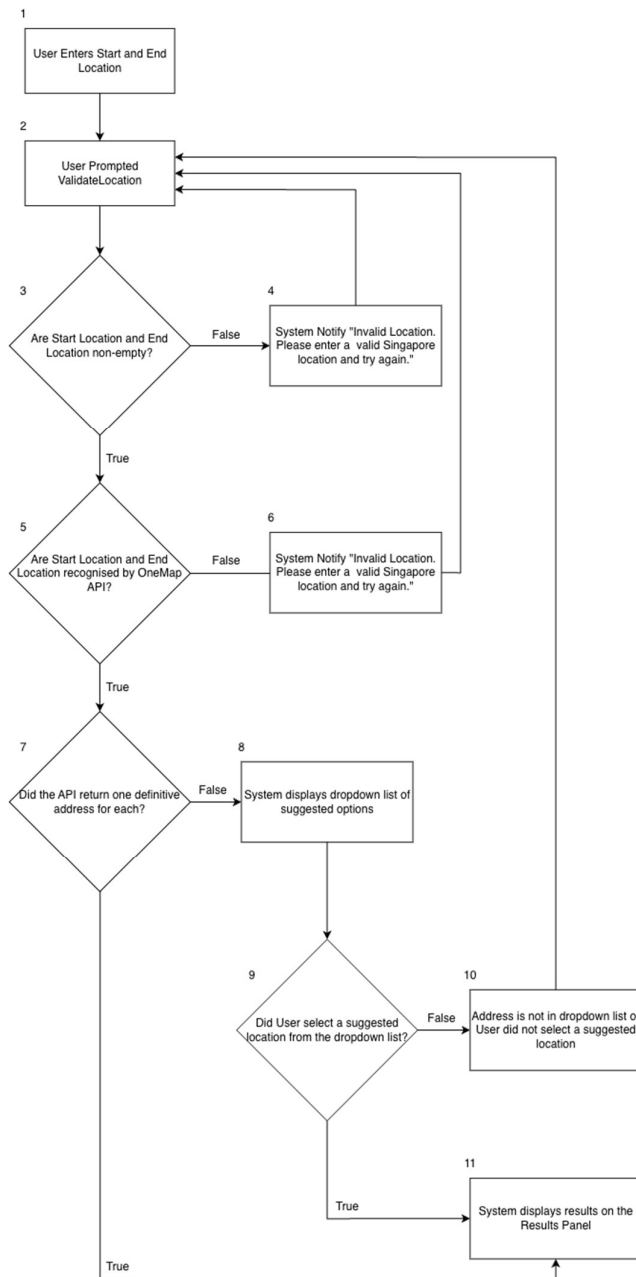
Login(Email, Password)

No.	Test Input	Expected Output	Actual Output	Pass?
Baseline	Email = "Admin1@gmail.com" Password = "12341234!"	"Log in successfully"	"Log in successfully"	Yes
1	Email = "" Password = "12341234!"	"Please enter your email and password!"	"Please enter your email and password!"	Yes
2	Email = "Admin1" Password = "12341234!"	"Please include an '@' at the email address."	"Please include an '@' at the email address."	Yes
3	Email = "Admin1@" Password = "12341234!"	"Please enter a part following '@'."	"Please enter a part following '@'."	Yes

4	Email = "Admin1@gmail.com" Password = ""	"Please enter your email and password!"	"Please enter your email and password!"	Yes
5	Email = "someone1@gmail.com" Password = "12341234!"	"Invalid login Credentials"	"Invalid login Credentials"	Yes
6	(5th Attempt incorrect) Email = "someone1@gmail.com" Password = "12341234!"	"Account is locked, please check your email to reset your password~"	"Account is locked, please check your email to reset your password~"	Yes

2. Validate Location

2.1 Control Flow Graph



2.2 Basic Path Testing

Cyclomatic Complexity = | decision points | + 1 = 4 + 1 = 5

Basis paths

1. Baseline Path 1: 1, 2, 3, 5, 7, 11
2. Baseline Path 2: 1, 2, 3, 4, 2
3. Baseline Path 3: 1, 2, 3, 5, 6, 2
4. Baseline Path 4: 1, 2, 3, 5, 7, 8, 9, 11
5. Baseline Path 5: 1, 2, 3, 5, 7, 8, 9, 10, 2

Explanation of basis paths

Path 1: User prompted to input start and end location. Start and end location are definitive addresses. The application will display the results on the Results Panel.

Path 2: User prompted to input start and end location. Start location is empty. User is prompted to input the address again.

Path 3: User prompted to input start and end location. Start location is not recognised by OneMap API. User is prompted to input the address again.

Path 4: User prompted to input start and end location. Start and End location are recognised by OneMap API but are not definitive addresses. Application displays a dropdown list of suggested locations. User selects a suggested location from the dropdown list. System displays results on the Results Panel

Path 5: User prompted to input start and end location. Start and End location are recognised by OneMap API but are not definitive addresses. Application displays a dropdown list of suggested locations. User does not select a location from the suggested dropdown list. User is prompted to address again.

2.3 Test Cases and Results

Location(Start, End)

No.	Test Input	Expected Output	Actual Output	Pass?
1	(Valid) Start Location : "50 Nanyang Ave, Singapore 639798" (Valid) End Location : " 301 Boon Lay Wy, Singapore 649846"	Location Validation Successful	Location Validation Successful	Yes
2	(Valid) Start Location : "50 Nanyang Ave, Singapore 639798" (Invalid) End Location : " "	System Notify "Invalid location. Please enter a	System Notify "Invalid location. Please enter a	Yes

		valid Singapore location and try again."	valid Singapore location and try again."	
3	(Invalid) Start Location : " " (Valid) End Location : " 301 Boon Lay Wy, Singapore 649846"	System Notify "Invalid location. Please enter a valid Singapore location and try again."	System Notify "Invalid location. Please enter a valid Singapore location and try again."	Yes
4	(Invalid) Start Location : " " (Invalid) End Location : " "	System Notify "Invalid location. Please enter a valid Singapore location and try again."	System Notify "Invalid location. Please enter a valid Singapore location and try again."	Yes
5	(Valid) Start Location : "50 Nanyang Ave, Singapore 639798" (Invalid) End Location : "Ochard"	End Location = "Orchard Road" Location Validation Successful	End Location = "Orchard Road" Location Validation Successful	Yes
6	(Invalid) Start Location : "Ochard" (Valid) End Location : " 301 Boon Lay Wy, Singapore 649846"	Start Location = "Orchard Road" Location Validation Successful	Start Location = "Orchard Road" Location Validation Successful	Yes